1.19 Graph Sketching: Quadratic Rational Functions

Worked Solutions

1 a)

No vertical asymptotes as $x^2 + 3 \neq 0$

As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$

So horizontal asymptote at $y = 1$

Asymptotes: $y = 1$

1 b)

No vertical asymptotes as $x^2 + 7 \neq 0$

As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$

So horizontal asymptote at $y = 1$

Asymptotes: $y = 1$

1 c)

$x^2 = 2$ gives $x = \pm\sqrt{2}$

As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$

So horizontal asymptote at $y = 1$

Asymptotes: $y = 1$, $x = -\sqrt{2}$ and $x = \sqrt{2}$

1 d)

$x^2 + 3x = 0$ gives $x = 0$ or $x = -3$

As $x \to \infty$, $y \to 3$ and as $x \to -\infty$, $y \to 3$

So horizontal asymptote at $y = 3$

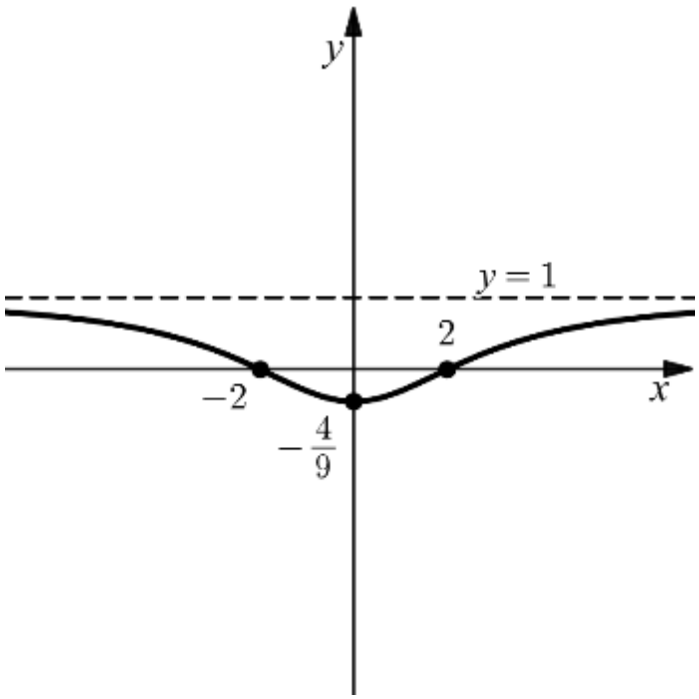
Asymptotes: $y = 3$, $x = 0$ and $x = -3$

2 a)

Asymptotes: $y = 1$

Intercepts: $(0, -\frac{4}{9})$, $(2, 0)$, $(-2, 0)$

When x is a large positive, $y < 1$, when x is a large negative, $y < 1$

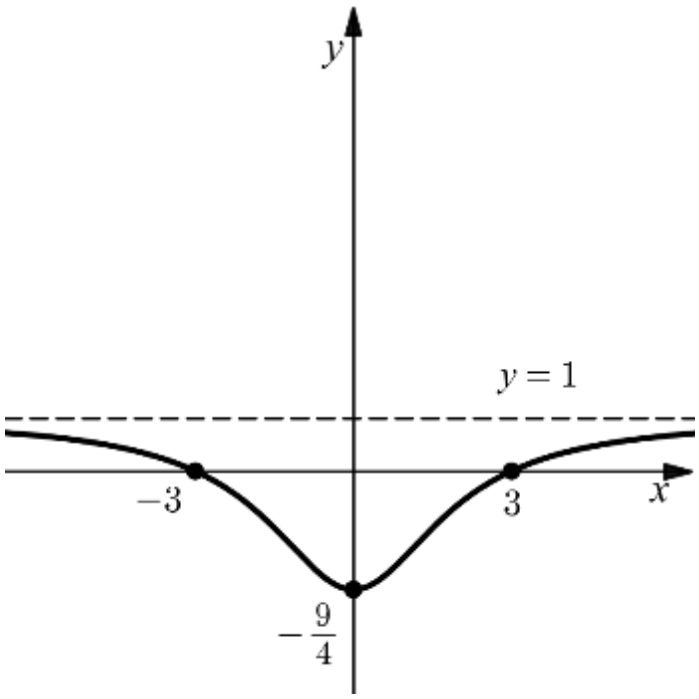


2 b)

Asymptotes: $y = 1$

Intercepts: $(0, -\frac{9}{4})$, $(3, 0)$, $(-3, 0)$

When x is a large positive, $y < 1$, when x is a large negative, $y < 1$

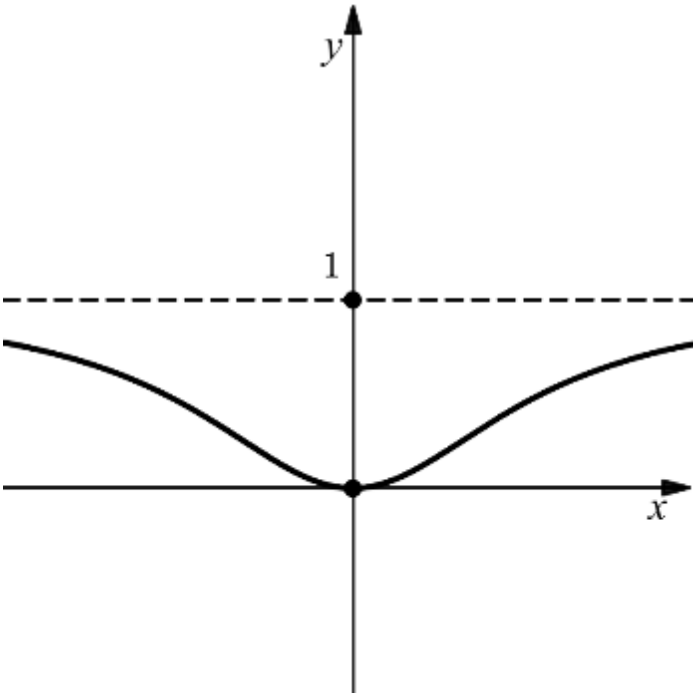


2 c)

Asymptotes: $y = 1$

Intercepts: $(0, 0)$

When x is a large positive, $y < 1$, when x is a large negative, $y < 1$



3 a)

$$\begin{aligned} 2(2x^2 + 3x + 9) &= 2x^2 - 3x + 23 \\ 4x^2 + 6x + 18 &= 2x^2 - 3x + 23 \\ 2x^2 + 9x - 5 &= 0 \\ (2x - 1)(x + 5) &= 0 \\ \therefore x &= \frac{1}{2} \text{ or } x = -5 \end{aligned}$$

3 b)

$$\begin{aligned} x^2 - 2x - 5 &= 2(x^2 - 4x + 2) \\ x^2 - 2x - 5 &= 2x^2 - 8x + 4 \\ x^2 - 6x + 9 &= 0 \\ (x - 3)^2 &= 0 \\ \therefore x &= 3 \end{aligned}$$

3 c)

$$\begin{aligned} 3(x^2 - 4x + 3) &= 4(x^2 - 3x + 2) \\ 3x^2 - 12x + 9 &= 4x^2 - 12x + 8 \\ x^2 - 1 &= 0 \\ (x - 1)(x + 1) &= 0 \end{aligned}$$

$x = 1$ or $x = -1$ but $x = 1$ gives 0 on the denominator

$$\therefore x = -1$$

Alternative Method:

$$\begin{aligned} \frac{x^2 - 4x + 3}{x^2 - 3x + 2} &= \frac{4}{3} \\ \Rightarrow \frac{(x-3)(x-1)}{(x-2)(x-1)} &= \frac{4}{3} \\ \Rightarrow \frac{x-3}{x-2} &= \frac{4}{3} \\ \Rightarrow 3x - 9 &= 4x - 8 \\ \therefore x &= -1 \end{aligned}$$

3 d)

$$\begin{aligned} -(x^2 + 4) &= 3(x^2 + 4x - 5) \\ -x^2 - 4 &= 3x^2 + 12x - 15 \\ 4x^2 + 12x - 11 &= 0 \\ \therefore x &= -\frac{3}{2} + \sqrt{5} \text{ or } x = -\frac{3}{2} - \sqrt{5} \end{aligned}$$

4 a)

Find the values for which y exists:

$$\begin{aligned} k = \frac{x^2 + 2x + 5}{x^2 + 2x - 3} &\Rightarrow k(x^2 + 2x - 3) = x^2 + 2x + 5 \\ (k-1)x^2 + (2k-2)x - (3k+5) &= 0 \end{aligned}$$

For k to exist: $(2k-2)^2 - 4(k-1)(-3k-5) \geq 0$

$$\begin{aligned} 4k^2 - 8k + 4 + 12k^2 + 8k - 20 &\geq 0 \\ 16k^2 - 16 &\geq 0 \\ (k-1)(k+1) &\geq 0 \text{ as req.} \end{aligned}$$

4 b)

$k = \pm 1$ from part a)

So $y > 1$ and $y \leq -1$

4 c)

P occurs when $y = -1$

5 a)

Find the values for which y exists:

$$k = \frac{x^2 - 8x + 4}{x^2 - 5x + 4} \Rightarrow k(x^2 - 5x + 4) = x^2 - 8x + 4$$

$$(k-1)x^2 + (8-5k)x + (4k-4) = 0$$

For k to exist: $(8-5k)^2 - 4(k-1)(4k-4) \geq 0$

$$25k^2 - 80k + 64 - 16k^2 + 32k - 16 \geq 0$$
$$9k^2 - 48k + 48 \geq 0$$

5 b)

$k = \frac{4}{3}$ or $k = 4$ from part a)

So $y \leq \frac{4}{3}$ and $y \geq 4$

5 c)

P occurs when $y = \frac{4}{3}$ and Q occurs when $y = 4$

6 a)

Find the values for which y exists:

$$k = \frac{x^2 + x + 5}{x^2 + 2x - 8} \Rightarrow k(x^2 + 2x - 8) = x^2 + x + 5$$
$$(k-1)x^2 + (2k-1)x - (8k+5) = 0$$

For k to exist: $(2k-1)^2 - 4(k-1)(-8k-5) \geq 0$

$$4k^2 - 4k + 1 + 32k^2 - 12k - 20 \geq 0$$
$$36k^2 - 16k - 19 \geq 0 \text{ as req.}$$

6 b)

$k = \frac{2}{9} \pm \frac{\sqrt{187}}{18}$ from part a)

Since $k = \frac{2}{9} + \frac{\sqrt{187}}{18} \approx 0.981\ldots$ and $k = \frac{2}{9} - \frac{\sqrt{187}}{18} \approx -0.537\ldots$

So $y > \frac{2}{9} + \frac{\sqrt{187}}{18}$ and $y \leq \frac{2}{9} - \frac{\sqrt{187}}{18}$

6 c)

P occurs when $y = \frac{2}{9} - \frac{\sqrt{187}}{18}$

7 a)

Find the values for which y exists:

$$k = \frac{x^2 - 2x}{x^2 - 8x + 16} \Rightarrow k(x^2 - 8x + 16) = x^2 - 2x$$
$$(k-1)x^2 + (2-8k)x + 16k = 0$$

For k to exist: $(2-8k)^2 - 4(k-1)16k \geq 0$

$$64k^2 - 32k + 4 - 64k^2 + 64k \geq 0$$
$$32k + 4 \geq 0 \text{ as req.}$$

7 b)

$k = -\frac{1}{8}$ from part a)

$$y \geq -\frac{1}{8}$$

7 c)

Q occurs when $y = -\frac{1}{8}$

8 a)

$x^2 + 5 > 0$ for all real values, so is not equal to zero

8 b)

$y = 0$

8 c)

$$k = \frac{x+2}{x^2+5} \Rightarrow k(x^2+5) = x+2$$
$$kx^2 + 5k = x + 2$$
$$kx^2 - x + 5k - 2 = 0$$
$$kx^2 - x + (5k-2) = 0 \text{ as req.}$$

8 d)

For k to exist: $(-1)^2 - 4k(5k-2) \geq 0$

$$-20k^2 + 8k + 1 \geq 0$$

$k = -\frac{1}{10}$ or $k = \frac{1}{2}$

So $-\frac{1}{10} \leq k \leq \frac{1}{2}$

8 e)

$$-\frac{1}{10} = \frac{x+2}{x^2+5} \Rightarrow -\frac{1}{10}x^2 - x + (5 \times -\frac{1}{10} - 2) = 0$$
$$x^2 + 10x + 5 = (x+5)^2 = 0 \quad \therefore x = -5$$
$$\frac{1}{2} = \frac{x+2}{x^2+5} \Rightarrow \frac{1}{2}x^2 - x + (5 \times \frac{1}{2} - 2) = 0$$
$$x^2 - 2x + 1 = (x-1)^2 = 0 \quad \therefore x = 1$$

So $(-5, -\frac{1}{10})$ and $(1, \frac{1}{2})$

9 a)

Find the values for which k exists:

$$k = \frac{12x}{x^2+4} \Rightarrow k(x^2+4) = 12x$$
$$kx^2 - 12x + 4k = 0$$

For k to exist: $(-12)^2 - 16k^2 \geq 0$

$$144 - 16k^2 \geq 0$$
$$(k+3)(k-3) \geq 0$$

So $-3 \leq k \leq 3$

And so $-3 \leq \frac{12x}{x^2+4} \leq 3$

9 b)

Intercept: $(0, 0)$

Asymptote: $y = 0$

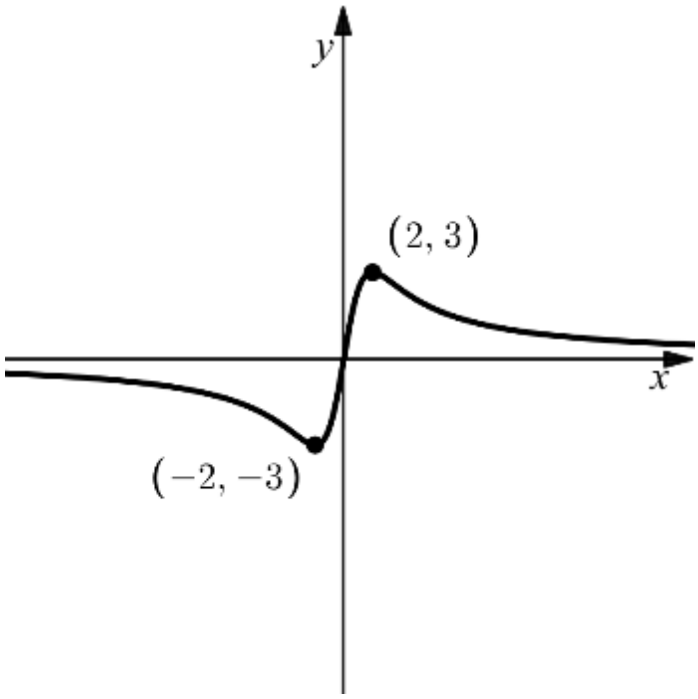
The graph exists for $-3 \leq y \leq 3$

When $y = 3$, $3 = \frac{12x}{x^2+4} \Rightarrow 3(x^2+4) = 12x \Rightarrow 3x^2 - 12x + 12 = 0$ so $x = 2$

Max is at $(2, 3)$

When $y = -3$, $-3 = \frac{12x}{x^2+4} \Rightarrow -3(x^2+4) = 12x \Rightarrow 3x^2 + 12x + 12 = 0$ so $x = -2$

Min is at $(-2, -3)$



9 c)

g(x) is the reciprocal of f(x), so g(x) can have the reciprocal values of f(x), that is, between $-\frac{1}{3}$ and $\frac{1}{3}$

However, g(x) is undefined for $x = 0$

10 a)

Asymptotes: $y = 1$, $x = -1$, $x = 1$

Intercepts: $(0, -4)$

When x is a large positive, $y > 1$, when x is a large negative, $y > 1$

Find the values for which y exists:

$$k = \frac{x^2+4}{x^2-1} \Rightarrow k(x^2-1) = x^2+4$$
$$(k-1)x^2 + (-k-4) = 0$$

For k to exist: $0^2 - 4(k-1)(-k-4) \geq 0$

$$4k^2 + 12k - 16 \geq 0$$

$k = -4$ or $k = 1$

So $k \leq -4$ and $k > 1$

When $y = -4$, $-4 = \frac{x^2+4}{x^2-1} \Rightarrow -4(x^2-1) = x^2+4 \Rightarrow 0 = 5x^2$ so $x = 0$

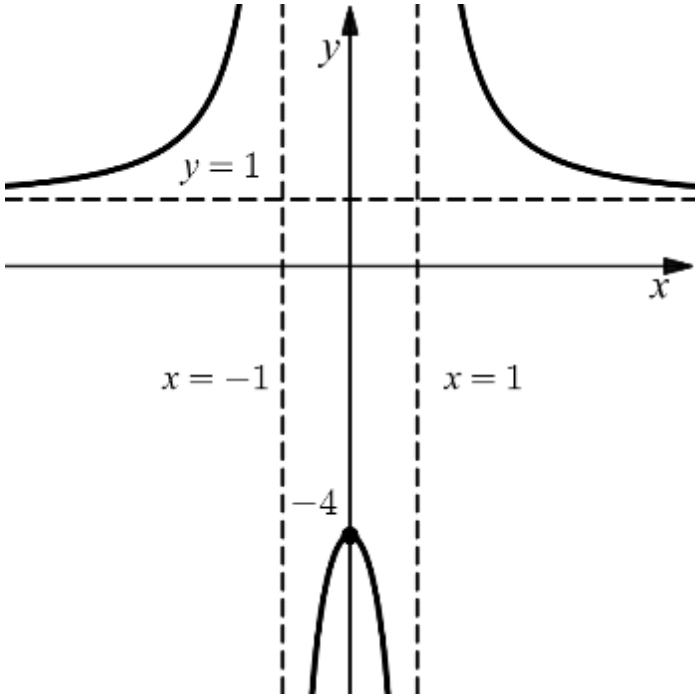
Local max is at $(0, -4)$

At $x = -1^-$, y is large positive

At $x = -1^+$, y is large negative

At $x = 1^-$, y is large negative

At $x = 1^+$, y is large positive



10 b)

Asymptotes: $y = 1$, $x = -4$, $x = 0$

$x^2 - 4x + 4 = 0 \Rightarrow x = 2$ so intercepts: $(2, 0)$

When $x \to \infty$, $y < 1$, when $x \to -\infty$, $y > 1$

Find the values for which y exists:

$$k = \frac{x^2-4x+4}{x^2+4x} \Rightarrow k(x^2+4x) = x^2-4x+4$$
$$(k-1)x^2 + (4k+4)x - 4 = 0$$

For k to exist: $(4k+4)^2 - 4(k-1)(-4) \geq 0$

$$16k^2 + 32k + 16 + 16k - 16 \geq 0$$
$$16k^2 + 48k \geq 0$$

$k = -3$ or $k = 0$

So $k \leq -3$ and $k \geq 0$

$$\text{When } y = -3, \quad -3 = \frac{x^2 - 4x + 4}{x^2 + 4x} \Rightarrow -3(x^2 + 4x) = x^2 - 4x + 4$$
$$\Rightarrow -3x^2 - 12x = x^2 - 4x + 4$$
$$\Rightarrow 0 = 4x^2 + 8x + 4 = 4(x+1)^2$$

so $x = -1$

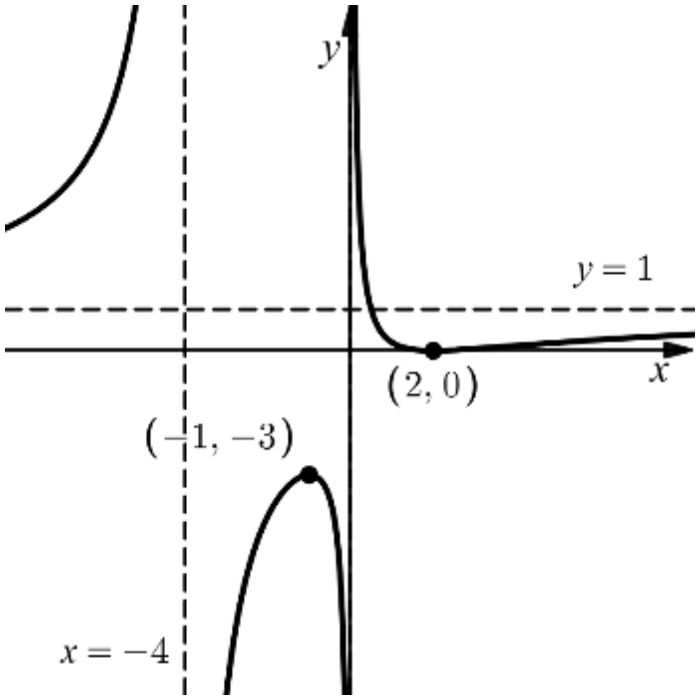
Local max is at $(-1, -3)$

At $x = -1^-$, y is large positive

At $x = -1^+$, y is large negative

At $x = 0^-$, y is large negative

At $x = 0^+$, y is large positive



10 c)

$x^2 + 2x + 2 = 0$ has no real solutions, so no vertical asymptotes.

Asymptotes: $y = 1$

$x^2 + 8x + 16 = 0 \Rightarrow x = -4$ so intercepts: $(-4, 0)$ and $(0, 8)$

When $x \to \infty$, $y > 1$, when $x \to -\infty$, $y < 1$

Find the values for which y exists:

$$k = \frac{x^2 + 8x + 16}{x^2 + 2x + 2} \Rightarrow k(x^2 + 2x + 2) = x^2 + 8x + 16$$

$$(k-1)x^2 + (2k-8)x + (2k-16) = 0$$

For k to exist: $(2k-8)^2 - 4(k-1)(2k-16) \geq 0$

$$4k^2 - 32k + 64 - 8k^2 + 72k - 64 \geq 0$$
$$-4k^2 + 40k \geq 0$$

$k = 0$ or $k = 10$

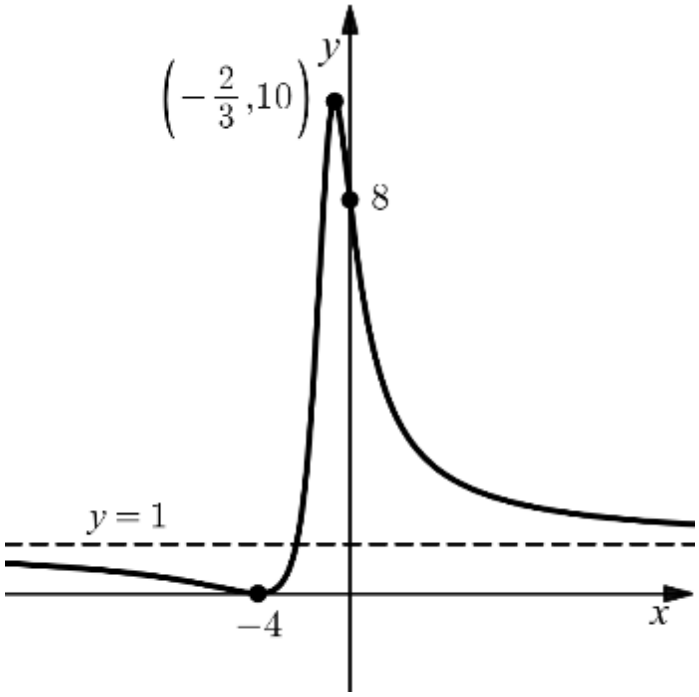
So $k \leq 10$ and $k \geq 0$

$$\text{When } y = 10, \quad 10 = \frac{x^2+8x+16}{x^2+2x+2} \Rightarrow 10(x^2+2x+2) = x^2+8x+16$$
$$\Rightarrow 9x^2 + 12x + 4 = 0 \text{ so } x = -\frac{2}{3}$$

Local max is at $(-\frac{2}{3}, 10)$

When $y = 0$, $0 = \frac{x^2+8x+16}{x^2+2x+2} \Rightarrow x = -4$

Local min is at $(-4, 0)$



10 d)

$x^2 + 2 = 0$ has no real solutions, so no vertical asymptotes.

Asymptotes: $y = 1$

$x^2 + 4x = 0 \Rightarrow x = -4$ so intercepts: $(-4, 0)$ and $(0, 0)$

When $x \to \infty$, $y > 1$, when $x \to -\infty$, $y < 1$

Find the values for which y exists:

$$k = \frac{x^2+4x}{x^2+2} \Rightarrow k(x^2+2) = x^2 + 4x$$
$$(k-1)x^2 - 4x + 2k = 0$$

For k to exist: $(-4)^2 - 4(k-1)2k \geq 0$

$$16 + 8k - 8k^2 \geq 0$$

$k = -1$ or $k = 2$

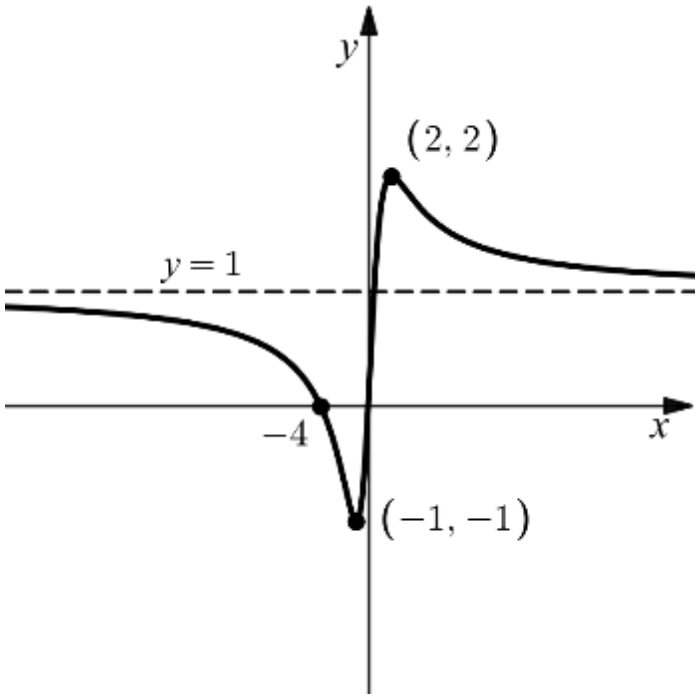
So $k \leq 2$ and $k \geq -1$

When $y = 2$, $2 = \frac{x^2+4x}{x^2+2} \Rightarrow 2(x^2+2) = x^2 + 4x \Rightarrow x^2 - 4x + 4 = 0$ so $x = 2$

Local max is at $(2, 2)$

When $y = -1$, $-1 = \frac{x^2+4x}{x^2+2} \Rightarrow -(x^2+2) = x^2 + 4x \Rightarrow 2x^2 + 4x + 2 = 0$ so $x = -1$

Local min is at $(-1, 1)$



10 e)

Asymptotes: $y = 1$, $x = 0$, $x = -4$

$x^2 - 4x - 5 = 0 \Rightarrow x = 5$, $x = -1$ so intercepts: $(5, 0)$ and $(-1, 0)$

No vertical intercept.

When $x \to \infty$, $y < 1$, when $x \to -\infty$, $y > 1$

Find the values for which y exists:

$$k = \frac{x^2 - 4x - 5}{x^2 + 4x} \Rightarrow k(x^2 + 4x) = x^2 - 4x - 5$$
$$(k-1)x^2 + (4k+4)x + 5 = 0$$

For k to exist: $(4k+4)^2 - 4(k-1)5 \geq 0$

$$16k^2 + 32k + 16 - 20k + 20 \geq 0$$
$$16k^2 + 12k + 36 \geq 0$$

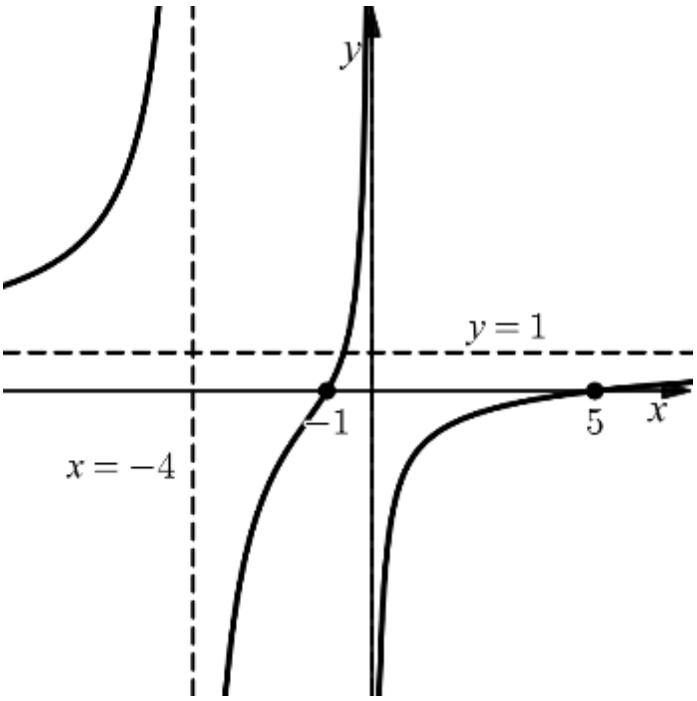
Which is true for all values of k, so y exists for all values (except $y = 1$).

At $x = -4^-$, y is large positive

At $x = -4^+$, y is large negative

At $x = 0^-$, y is large positive

At $x = 0^+$, y is large negative



10 f)

$x^2 - 4x + 3 = 0 \Rightarrow x = 1,\ x = 3$

Asymptotes: $y = 1$, $x = 1$, $x = 3$

$x^2 - 4x + 2 = 0 \Rightarrow x = 2 - \sqrt{2},\ x = 2 + \sqrt{2}$ so intercepts: $(2 - \sqrt{2}, 0)$ and $(2 + \sqrt{2}, 0)$

Vertical intercept at $(0, \frac{2}{3})$

When $x \to \infty$, $y < 1$, when $x \to -\infty$, $y > 1$

Find the values for which y exists:

$$k = \frac{x^2 - 4x + 2}{x^2 - 4x + 3} \Rightarrow k(x^2 - 4x + 3) = x^2 - 4x + 2$$
$$(k-1)x^2 + (4 - 4k)x + (3k - 2) = 0$$

For k to exist: $(4 - 4k)^2 - 4(k - 1)(3k - 2) \geq 0$

$$16k^2 - 32k + 16 - 12k^2 + 20k - 8 \geq 0$$
$$4k^2 - 12k + 8 \geq 0$$

$k = 1$ or $k = 2$

So $k < 1$ and $k \geq 2$

$$\text{When } y = 2, \quad 2 = \frac{x^2 - 4x + 2}{x^2 - 4x + 3} \Rightarrow 2(x^2 - 4x + 3) = x^2 - 4x + 2$$
$$\Rightarrow x^2 - 4x + 4 = 0 \text{ so } x = 2$$

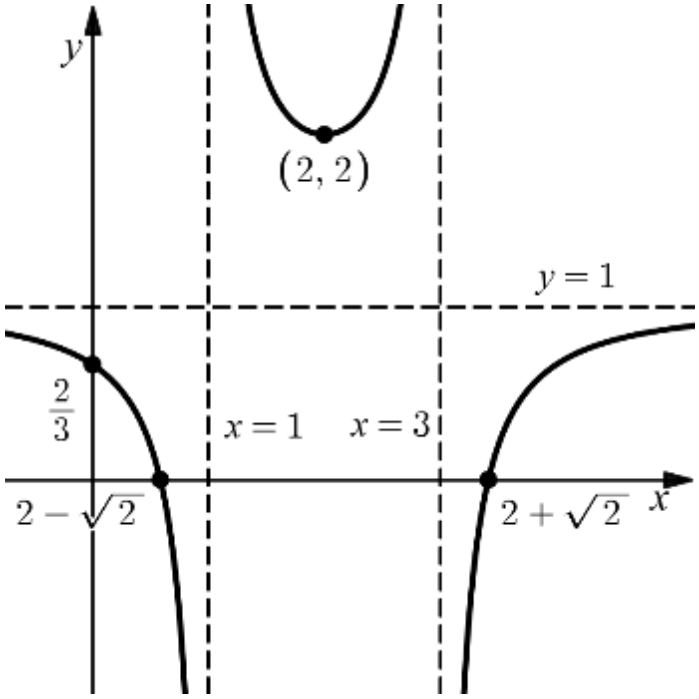
Local min is at $(2, 2)$

At $x = 1^-$, y is large negative

At $x = 1^+$, y is large positive

At $x = 3^-$, y is large positive

At $x = 3^+$, y is large negative



11 a)

Horizontal asymptote: $y = 2$

Vertical asymptote: none as $x^2 + 2x + 5 > 0$

$$\therefore y = 2$$

11 b)

Horizontal asymptote: $y = 3$

Vertical asymptote: none as $x^2 + 2x + 3 > 0$

$$\therefore y = 3$$

11 c)

Horizontal asymptote: $y = \frac{1}{4}$

Vertical asymptote: $4x^2 - 8x + 4 = 0$ so $x = 1$

$$\therefore y = \frac{1}{4}, \quad x = 1$$

11 d)

Horizontal asymptote: $y = \frac{5}{4}$

Vertical asymptote: $2x^2 + 7x - 15 = 0$ so $x = -5$ or $x = \frac{3}{2}$

$$\therefore y = \frac{5}{4}, \quad x = -5, \quad x = \frac{3}{2}$$

12)

$y = a$, $x = \sqrt{d}$, $x = -\sqrt{d}$

13 a)

$y = 4$

13 b)

Find the values for which y exists:

$$k = \frac{4x^2 + 4x + 11}{x^2 - x + 1} \Rightarrow k(x^2 - x + 1) = 4x^2 + 4x + 11$$
$$(k-4)x^2 + (-k-4)x + (k-11) = 0$$

For k to exist: $(-k-4)^2 - 4(k-4)(k-11) \geq 0$

$$\begin{aligned} k^2 + 8k + 16 - 4k^2 + 60k - 176 &\geq 0 \\ -3k^2 + 68k - 160 &\geq 0 \\ 3k^2 - 68k + 160 &\leq 0 \\ (3k-8)(k-20) &\leq 0 \text{ as req.} \end{aligned}$$

13 c)

The graph exists for $\frac{8}{3} \leq y \leq 20$

$$\text{When } y = 20, \quad 20 = \frac{4x^2 + 4x + 11}{x^2 - x + 1} \Rightarrow 20(x^2 - x + 1) = 4x^2 + 4x + 11$$
$$\Rightarrow 16x^2 - 24x + 9 = 0 \text{ so } x = \frac{3}{4}$$

Max is at $(\frac{3}{4}, 20)$

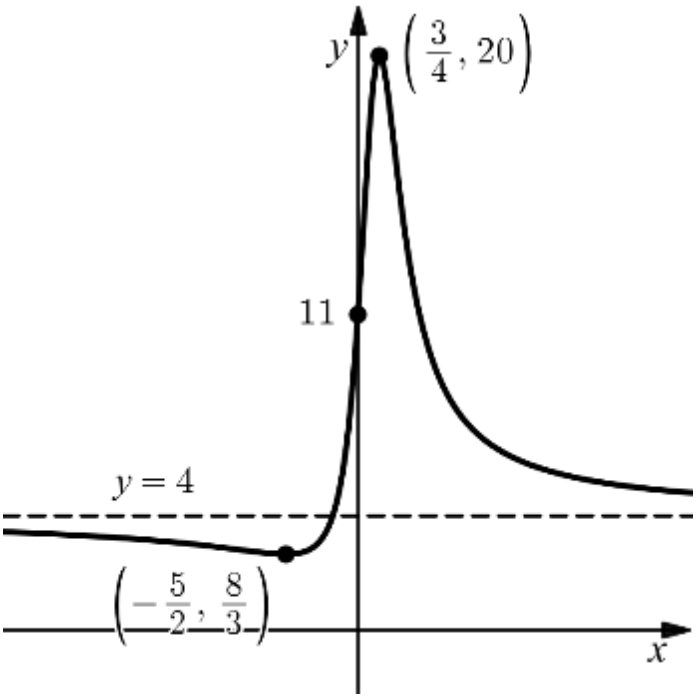
$$\text{When } y = \frac{8}{3}, \quad \frac{8}{3} = \frac{4x^2 + 4x + 11}{x^2 - x + 1} \Rightarrow \frac{8}{3}(x^2 - x + 1) = 4x^2 + 4x + 11$$
$$\Rightarrow 8x^2 - 8x + 8 = 12x^2 + 12x + 33$$
$$\Rightarrow 4x^2 + 20x + 25 = 0 \text{ so } x = -\frac{5}{2}$$

Min is at $(-\frac{5}{2}, \frac{8}{3})$

$\therefore$ Max at $(\frac{3}{4}, 20)$ and min at $(-\frac{5}{2}, \frac{8}{3})$ as the graph exists for $\frac{8}{3} \leq y \leq 20$

13 d)

Intercept at $(0, 11)$



14 a)

$p = 2$

14 b)

$x = 2$, $x = -2$, $y = 2$

14 c)

Find the values for which y exists:

$$k = \frac{2x^2 - 4x + 9}{x^2 - 4} \Rightarrow k(x^2 - 4) = 2x^2 - 4x + 9$$
$$(k-2)x^2 + 4x + (-4k - 9) = 0$$

For k to exist: $4^2 - 4(k-2)(-4k-9) \geq 0$

$$16 + 16k^2 + 4k - 72 \geq 0$$
$$4k^2 + k - 14 \geq 0$$
$$(4k - 7)(k + 2) \geq 0 \text{ as req.}$$

14 d)

The graph exists for $y \leq -2$ and $y \geq \frac{7}{4}$ (which is below the asymptote)

$$\text{When } y = -2, \quad -2 = \frac{2x^2 - 4x + 9}{x^2 - 4} \Rightarrow -2(x^2 - 4) = 2x^2 - 4x + 9$$
$$\Rightarrow 4x^2 - 4x + 1 = 0 \text{ so } x = \frac{1}{2}$$

Local max is at $(\frac{1}{2}, -2)$

$$\text{When } y = \frac{7}{4}, \quad \frac{7}{4} = \frac{2x^2 - 4x + 9}{x^2 - 4} \Rightarrow \frac{7}{4}(x^2 - 4) = 2x^2 - 4x + 9$$
$$\Rightarrow 7x^2 - 28 = 8x^2 - 16x + 36$$
$$\Rightarrow x^2 - 16x + 64 = 0 \text{ so } x = 8$$

Local min is at $(8, \frac{7}{4})$

$\therefore$ Local max is at $(\frac{1}{2}, -2)$ and local min is at $(8, \frac{7}{4})$

14 e)

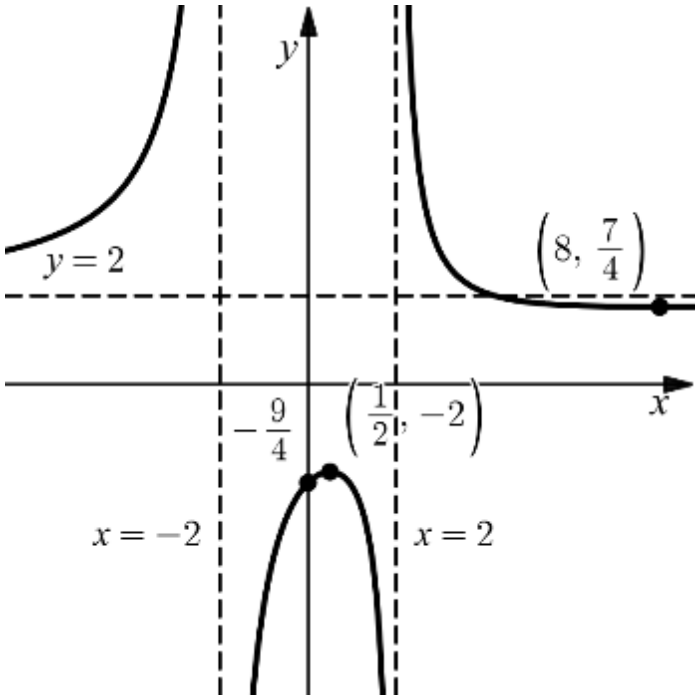
Intercept at $(0, -\frac{9}{4})$

At $x = -2^-$, y is large positive

At $x = -2^+$, y is large negative

At $x = 2^-$, y is large negative

At $x = 2^+$, y is large positive



15 a)

$y = \frac{1}{3}$

15 b)

Find the values for which y exists:

$$k = \frac{x^2 - 8x + 1}{3x^2 + 1} \Rightarrow k(3x^2 + 1) = x^2 - 8x + 1$$
$$(3k - 1)x^2 + 8x + (k - 1) = 0$$

For k to exist: $8^2 - 4(3k - 1)(k - 1) \geq 0$

$$64 - 12k^2 + 16k - 4 \geq 0$$

$$-12k^2 + 16k + 60 \geq 0$$
$$3k^2 - 4k - 15 \leq 0$$
$$(3k+5)(k-3) \leq 0 \text{ as req.}$$

15 c)

The graph exists for $-\frac{5}{3} \leq y \leq 3$

$$\text{When } y = 3, \quad 3 = \frac{x^2 - 8x + 1}{3x^2 + 1} \Rightarrow 3(3x^2+1) = x^2 - 8x + 1$$
$$\Rightarrow 8x^2 + 8x + 2 = 0 \text{ so } x = -\frac{1}{2}$$

Max is at $(-\frac{1}{2}, 3)$

$$\text{When } y = -\frac{5}{3}, \quad -\frac{5}{3} = \frac{x^2 - 8x + 1}{3x^2 + 1} \Rightarrow -\frac{5}{3}(3x^2+1) = x^2 - 8x + 1$$
$$\Rightarrow -15x^2 - 5 = 3x^2 - 24x + 3$$
$$\Rightarrow 18x^2 - 24x + 8 = 0 \text{ so } x = \frac{2}{3}$$

Min is at $(\frac{2}{3}, -\frac{5}{3})$

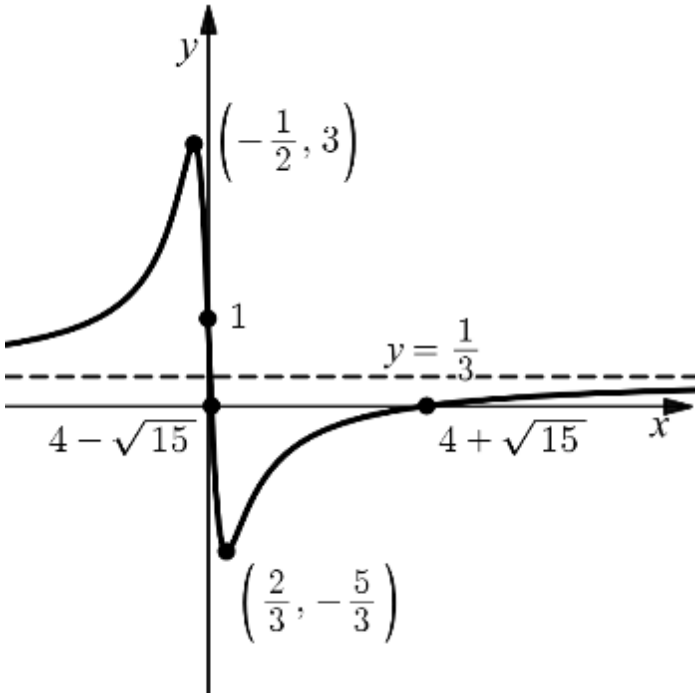
$\therefore$ Max is at $(-\frac{1}{2}, 3)$ and min is at $(\frac{2}{3}, -\frac{5}{3})$

15 d)

Intercept at $(0, 1)$

$x^2 - 8x + 1 = 0$ so $x = 4 \pm \sqrt{15}$

But since the curve tends to $y = \frac{1}{3}$ it only crosses the curve once and it is for a larger x-value than $x = \frac{2}{3}$



16 a)

Asymptote: $y = 0$

Find the values for which y exists:

$$k = \frac{1}{2x^2+3} \Rightarrow k(2x^2+3) = 1$$
$$2kx^2 + (3k-1) = 0$$

For k to exist: $0^2 - 4(2k)(3k-1) \geq 0$

$$-24k^2 + 8k \geq 0$$
$$8k(3k-1) \leq 0$$

The graph exists for $0 < y \leq \frac{1}{3}$

$$\text{When } y = \frac{1}{3}, \quad \frac{1}{3} = \frac{1}{2x^2+3} \Rightarrow \frac{1}{3}(2x^2+3) = 1$$
$$\Rightarrow 2x^2 = 0 \text{ so } x = 0$$

Max is at $(0, \frac{1}{3})$

$\therefore$ Max is at $(0, \frac{1}{3})$ and asymptote is $y = 0$

16 b)

Asymptote: $y = 0$

Find the values for which y exists:

$$k = \frac{1}{3(x^2+2x+2)} \Rightarrow 3k(x^2+2x+2) = 1$$
$$3kx^2 + 6kx + (6k-1) = 0$$

For k to exist: $(6k)^2 - 4(3k)(6k-1) \geq 0$

$$36k^2 - 72k^2 + 12k \geq 0$$
$$3k^2 - k \leq 0$$
$$k(3k-1) \leq 0$$

The graph exists for $0 < y \leq \frac{1}{3}$

$$\text{When } y = \frac{1}{3}, \quad \frac{1}{3} = \frac{1}{3(x^2+2x+2)} \Rightarrow x^2+2x+2 = 1$$
$$\Rightarrow x^2 + 2x + 1 = 0 \text{ so } x = -1$$

Max is at $(-1, \frac{1}{3})$

$\therefore$ Max is at $(-1, \frac{1}{3})$ and asymptote is $y = 0$

17)

Since g(x) is the reciprocal of f(x) it will become a minimum point as long as $b \neq 0$

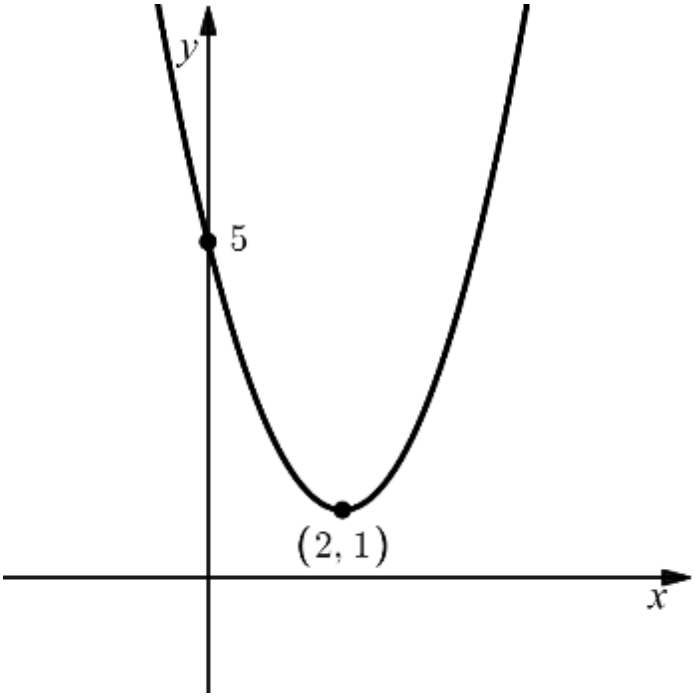
18 a)

$x^2 - 4x + 5 = (x-2)^2 + 1$

So minimum point is at $(2, 1)$

18 b)

Intercept: $(0, 5)$



18 c)

Reciprocal of previous minimum:

$$\text{When } y = \frac{1}{1} = 1, \quad 1 = \frac{1}{x^2 - 4x + 5} \Rightarrow x^2 - 4x + 5 = 1$$
$$\Rightarrow x^2 - 4x + 4 = 0 \text{ so } x = 2$$

$\therefore$ Max is at $(2, 1)$

Alternative Method:

Find the values for which y exists:

$$k = \frac{1}{x^2 - 4x + 5} \Rightarrow k(x^2 - 4x + 5) = 1$$
$$kx^2 - 4kx + (5k - 1) = 0$$

For k to exist: $(-4k)^2 - 4k(5k - 1) \geq 0$

$$16k^2 - 20k^2 + 4k \geq 0$$
$$k^2 - k \leq 0$$
$$k(k - 1) \leq 0$$

The graph exists for $0 < y \leq 1$ (since asymptote at $y = 0$)

$$\text{When } y = 1, \quad 1 = \frac{1}{x^2 - 4x + 5} \Rightarrow x^2 - 4x + 5 = 1$$
$$\Rightarrow x^2 - 4x + 4 = 0 \text{ so } x = 2 \quad \therefore \text{ Max is at } (2, 1)$$

18 d)

Since both graphs have the same stationary point, the tangent line is $y = 1$

19 a)

$$k = \frac{3x^2 - 8x + 7}{x^2 + 4x - 6} \Rightarrow k(x^2 + 4x - 6) = 3x^2 - 8x + 7$$
$$(k-3)x^2 + (4k+8)x + (-6k-7) = 0$$

For k to exist: $(4k+8)^2 - 4(k-3)(-6k-7) \geq 0$

$$16k^2 + 64k + 64 + 24k^2 - 44k - 84 \geq 0$$
$$40k^2 + 20k - 20 \geq 0$$
$$(2k-1)(k+1) \geq 0 \text{ as req.}$$

19 b)

$k \leq -1$ or $k \geq \frac{1}{2}$

The graph exists for $y \leq -1$ and $y \geq \frac{1}{2}$

$$\text{When } y = -1, \quad -1 = \frac{3x^2 - 8x + 7}{x^2 + 4x - 6} \Rightarrow -(x^2 + 4x - 6) = 3x^2 - 8x + 7$$
$$\Rightarrow 4x^2 - 4x + 1 = 0 \text{ so } x = \frac{1}{2}$$

Local max is at $(\frac{1}{2}, -1)$

$$\text{When } y = \frac{1}{2}, \quad \frac{1}{2} = \frac{3x^2 - 8x + 7}{x^2 + 4x - 6} \Rightarrow \frac{1}{2}(x^2 + 4x - 6) = 3x^2 - 8x + 7$$
$$\Rightarrow x^2 + 4x - 6 = 6x^2 - 16x + 14$$
$$\Rightarrow 5x^2 - 20x + 20 = 0 \text{ so } x = 2$$

Local Min is at $(2, \frac{1}{2})$

$\therefore$ Local max is at $(\frac{1}{2}, -1)$ and local min is at $(2, \frac{1}{2})$

19 c)

$y = \frac{x^2+4x-6}{3x^2-8x+7}$

$3x^2 - 8x + 7 = 0$ has no real solutions (the solutions are $\frac{4}{3} \pm \frac{\sqrt{5}}{3}$i)

$\therefore$ no vertical asymptotes.

19 d)

Since f$(x) \geq \frac{1}{2}$ and f$(x) \leq -1$ then $-1 \leq \frac{1}{\text{f}(x)} \leq 2$

So minimum of $\frac{1}{\text{f}(x)}$ is when $\frac{1}{\text{f}(x)} = -1$

$$-1 = \frac{x^2 + 4x - 6}{3x^2 - 8x + 7} \Rightarrow -(3x^2 - 8x + 7) = x^2 + 4x - 6$$
$$\Rightarrow 4x^2 - 4x + 1 = 0 \text{ so } x = \frac{1}{2}$$

So it has a minimum at $(\frac{1}{2}, -1)$ which is the local maximum of f(x)

So the tangent line must be $y = -1$

20 a)

$y = -1$

20 b)

No solutions to $x^2 + kx + 2 = 0$

$$k^2 - 4 \times 2 < 0$$
$$(k - 2\sqrt{2})(k + 2\sqrt{2}) < 0$$

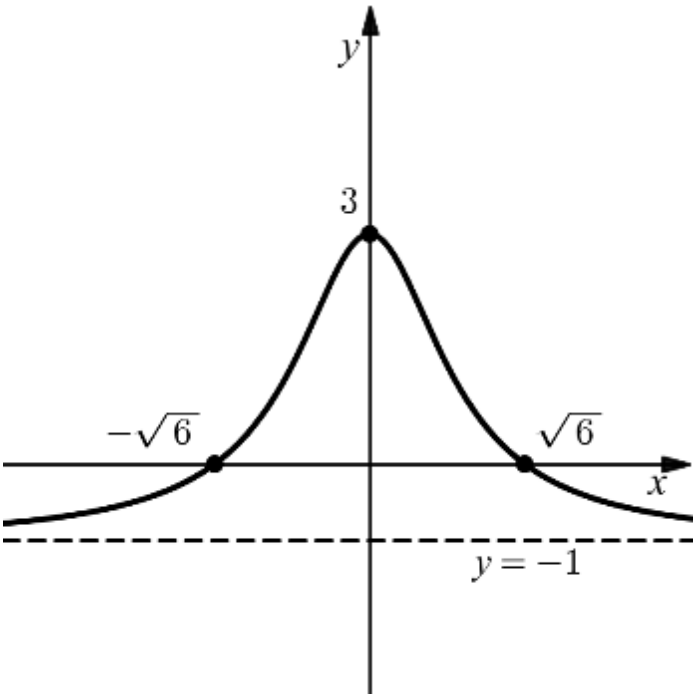
So $-2\sqrt{2} < k < 2\sqrt{2}$

20 c)

$(0, 3)$, $(\sqrt{6}, 0)$ and $(-\sqrt{6}, 0)$

20 d)

The curve is symmetrical about the y-axis when $k = 0$



21 a)

$a = 4$

21 b)

$$x^2 + bx - 8 = 0$$
$$(x - 4)(x + 2) = 0$$
$$x^2 - 2x - 8 = 0$$

So $b = -2$

21 c)

$x = -2$

21 d)

$y = \frac{4x^2-8x+8}{x^2-2x-8}$

$$(0, -1)$$

21 e)

Find the values for which y exists:

$$k = \frac{4x^2 - 8x + 8}{x^2 - 2x - 8} \Rightarrow k(x^2 - 2x - 8) = 4x^2 - 8x + 8$$
$$(k-4)x^2 + (8-2k)x + (-8k-8) = 0$$

For k to exist: $(8-2k)^2 - 4(k-4)(-8k-8) \geq 0$

$$4k^2 - 32k + 64 + 32k^2 - 96k - 128 \geq 0$$
$$36k^2 - 128k - 64 \geq 0$$
$$(9k-4)(k-4) \geq 0$$

The graph exists for $y \leq -\frac{4}{9}$ or $y > 4$

$$\text{When } y = -\frac{4}{9}, \quad -\frac{4}{9} = \frac{4x^2 - 8x + 8}{x^2 - 2x - 8} \Rightarrow -4(x^2 - 2x - 8) = 9(4x^2 - 8x + 8)$$
$$\Rightarrow x^2 - 2x + 1 = 0 \text{ so } x = 1$$

$\therefore$ Local max is at $(1, -\frac{4}{9})$

22 a)

$a = 2$

When $y = 0$, $x = -1 \pm \sqrt{\frac{3}{2}}$

$$2x^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 8c}}{4}$$

So $b = 4$

$$\frac{\sqrt{16 - 8c}}{4} = \sqrt{\frac{3}{2}} \Rightarrow \frac{16 - 8c}{16} = \frac{3}{2} \Rightarrow 16 - 8c = 24 \quad \therefore c = -1$$

When $x = 0$, $y = -1$ so $\frac{-1}{q} = -1 \quad \therefore q = 1$

$$y = \frac{2x^2 + 4x - 1}{x^2 + px - 1}$$

At $(-\frac{1}{2}, -2)$

$$-2 = \frac{2(-\frac{1}{2})^2 + 4(-\frac{1}{2}) - 1}{(-\frac{1}{2})^2 + p(-\frac{1}{2}) - 1}$$
$$\Rightarrow -2 = \frac{-\frac{5}{2}}{-\frac{1}{2}p - \frac{5}{4}}$$
$$\Rightarrow p + \frac{10}{4} = -\frac{5}{2} \quad \therefore p = 0$$
$$\mathrm{f}(x) = \frac{2x^2 + 4x - 1}{x^2 + 1}$$

22 b)

Find the values for which y exists:

$$k = \frac{2x^2 + 4x - 1}{x^2 + 1} \Rightarrow k(x^2 + 1) = 2x^2 + 4x - 1$$
$$(k-2)x^2 - 4x + (k+1) = 0$$

For k to exist: $(-4)^2 - 4(k-2)(k+1) \geq 0$

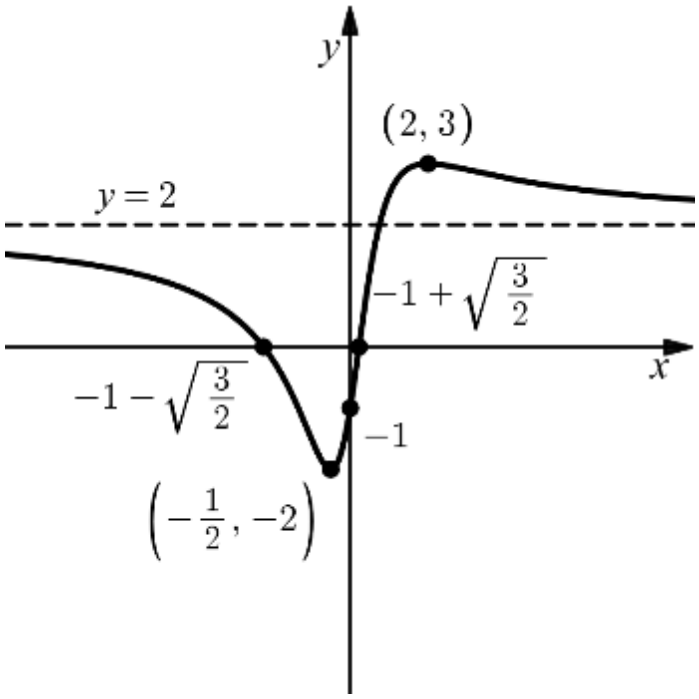
$$16 - 4k^2 + 4k + 8 \geq 0$$
$$k^2 - k - 6 \leq 0$$
$$(k+2)(k-3) \leq 0$$

The graph exists for $-2 \leq y \leq 3$

$$\text{When } y = 3, \quad 3 = \frac{2x^2 + 4x - 1}{x^2 + 1} \Rightarrow 3(x^2 + 1) = 2x^2 + 4x - 1$$
$$\Rightarrow x^2 - 4x + 4 = 0 \text{ so } x = 2$$

$\therefore$ Local max is at $(2, 3)$

22 c)



23 a)

$a = p$

23 b)

When $x = 0$, $y = 2$ so $\frac{c}{r} = 2$

$$\therefore c = 2r$$

23 c)

Since $a = 2$

When $y = 0$, $x = 2$ or $x = -\frac{3}{2}$

So the numerator is $k(x-2)(2x+3)$ is for some value of k

$$k(2x^2 - x - 6)$$

So $k = 1$

$$\mathrm{f}(x) = \frac{2x^2 - x - 6}{2x^2 + qx + r}$$

But since $c = 2r$ then $r = -3$

$$\mathrm{f}(x) = \frac{2x^2 - x - 6}{2x^2 + qx - 3}$$

When $x = -3$, $2x^2 + qx - 3 = 0$ so $18 - 3q - 3 = 0 \quad \therefore q = 5$

$$\mathrm{f}(x) = \frac{2x^2 - x - 6}{2x^2 + 5x - 3}$$

23 d)

$2x^2 + 5x - 3 = 0$ has solutions $x = -3$ and $x = \frac{1}{2}$

So the other asymptote is $x = \frac{1}{2}$

24)

Find the values for which y exists:

$$k = \frac{x^2 + p}{x^2 - q^2} \Rightarrow k(x^2 - q^2) = x^2 + p$$
$$(k-1)x^2 + (-kq^2 - p) = 0$$

For k to exist: $0^2 - 4(k-1)(-kq^2 - p) \geq 0$

$$\Rightarrow (k-1)(-kq^2 - p) \leq 0$$
$$\Rightarrow (k-1)(q^2k + p) \geq 0$$
$$\Rightarrow k \leq -\frac{p}{q^2} \text{ or } k \geq 1$$

The graph exists for $y \leq -\frac{p}{q^2}$ and $y \geq 1$

25 a)

No vertical asymptotes.

$$y = 1$$

25 b)

Find the values for which y exists:

$$k = \frac{x^2 - px + p}{x^2 + q} \Rightarrow k(x^2 + q) = x^2 - px + p$$
$$(k-1)x^2 + px + (kq - p) = 0$$

For k to exist: $p^2 - 4(k-1)(kq-p) \geq 0$

$$\begin{aligned} p^2 - 4qk^2 + 4(p+q)k - 4p &\geq 0 \\ 4qk^2 - 4(p+q)k + (4p - p^2) &\leq 0 \end{aligned}$$

Solving $4qk^2 - 4(p+q)k + (4p-p^2) = 0$ gives

$$\begin{aligned} k &= \frac{4(p+q) \pm \sqrt{(-4(p+q))^2 - 4(4q)(4p-p^2)}}{8q} \\ &= \frac{4(p+q) \pm \sqrt{16((p+q))^2 - 16q(4p-p^2)}}{8q} \\ &= \frac{4(p+q) \pm 4\sqrt{(p+q)^2 - q(4p-p^2)}}{8q} \\ &= \frac{p+q \pm \sqrt{p^2+2pq+q^2-4pq+p^2q}}{2q} \\ &= \frac{p+q \pm \sqrt{p^2-2pq+q^2+p^2q}}{2q} \end{aligned}$$

So the graph does not exist for

$$y < \frac{p+q-\sqrt{p^2-2pq+q^2+p^2q}}{2q}$$

and

$$y > \frac{p+q \pm \sqrt{p^2-2pq+q^2+p^2q}}{2q}$$